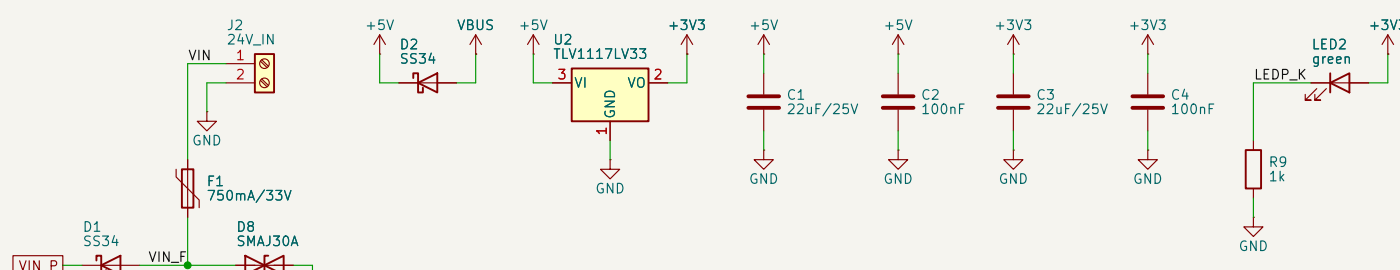
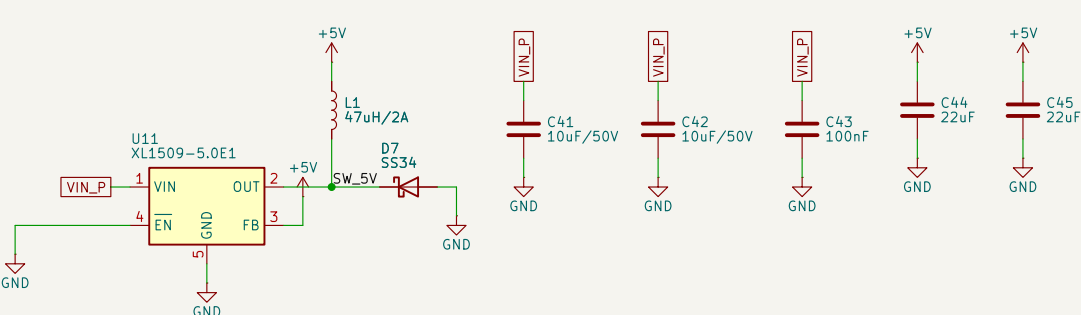


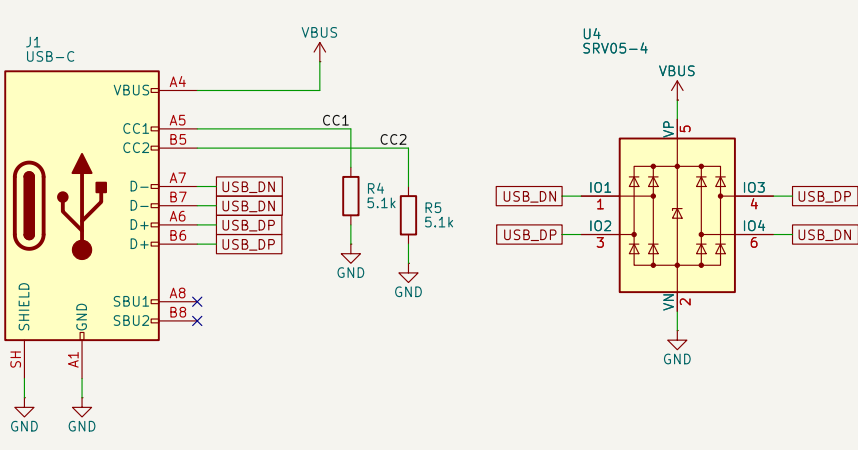
POWER IN: 24 VDC terminal or USB
F1+D8 clamp-and-blow at the entry; D1 blocks reverse polarity.
U11 (right) makes +5V; D2 ORs in USB VBUS; U2 drops it to +3V3



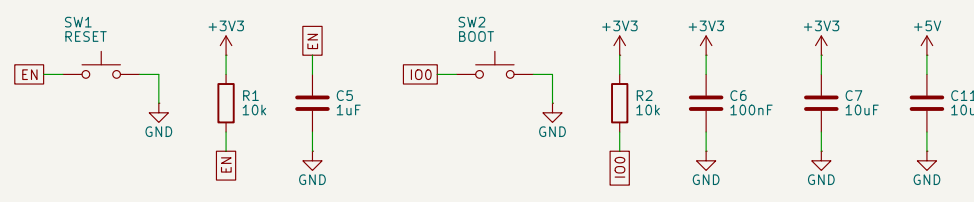
24 V -> 5 V BUCK: XL1509-5.0, fixed output
C41-C43 in, D7 catch, L1 47uH, C44/C45 out, F8 ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. Keep C41/C42 - U11 - D7 tight: that loop, not L1, is what radiates.



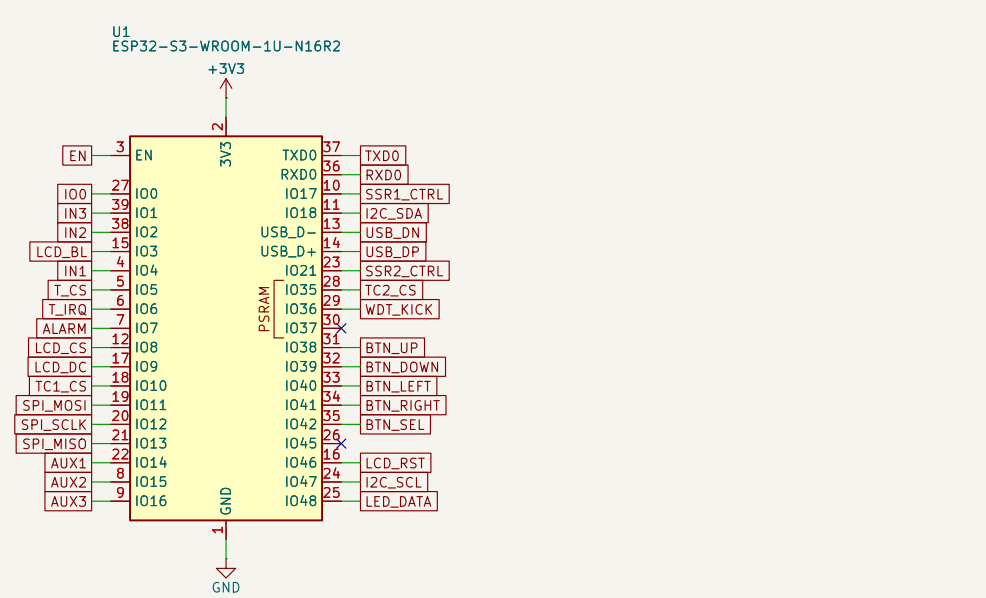
USB-C native USB flashing + ESD



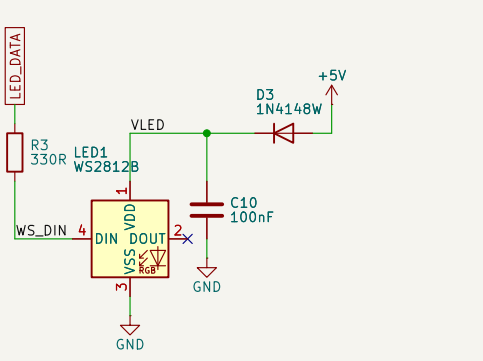
RESET / BOOT / DECOUPLING



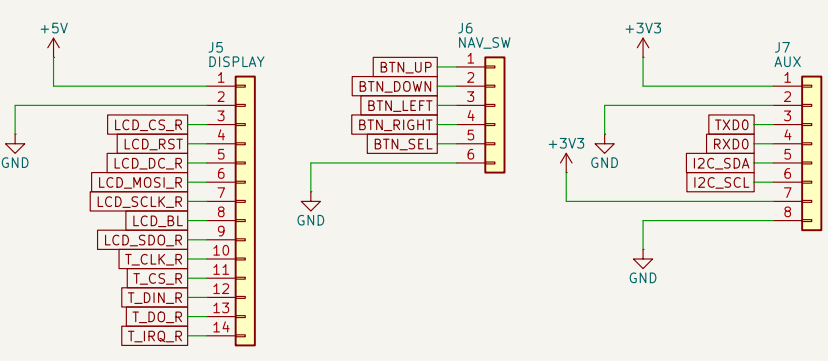
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



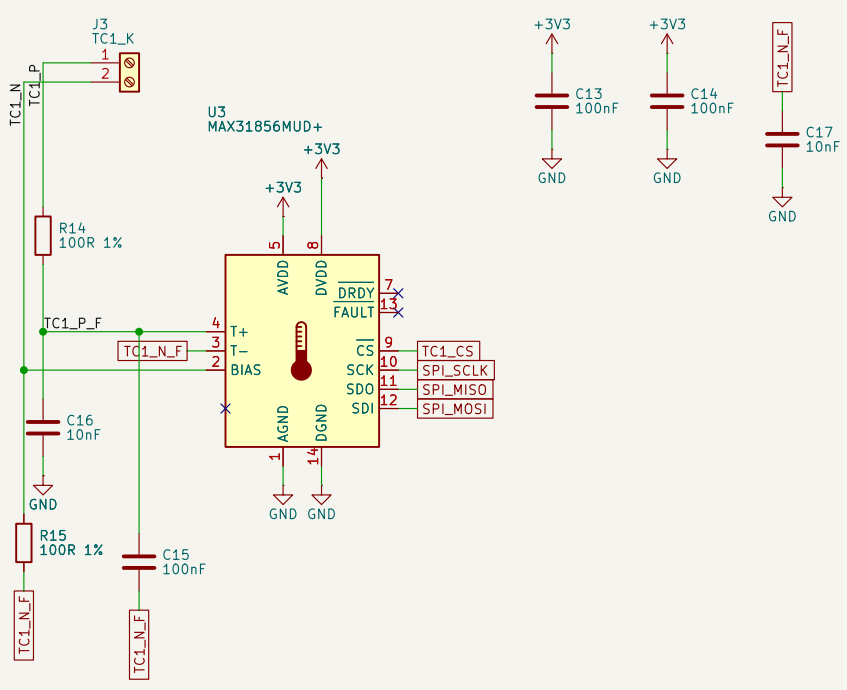
WS2812B STATUS LED
VDD dropped -4.6V for 3.3V data margin



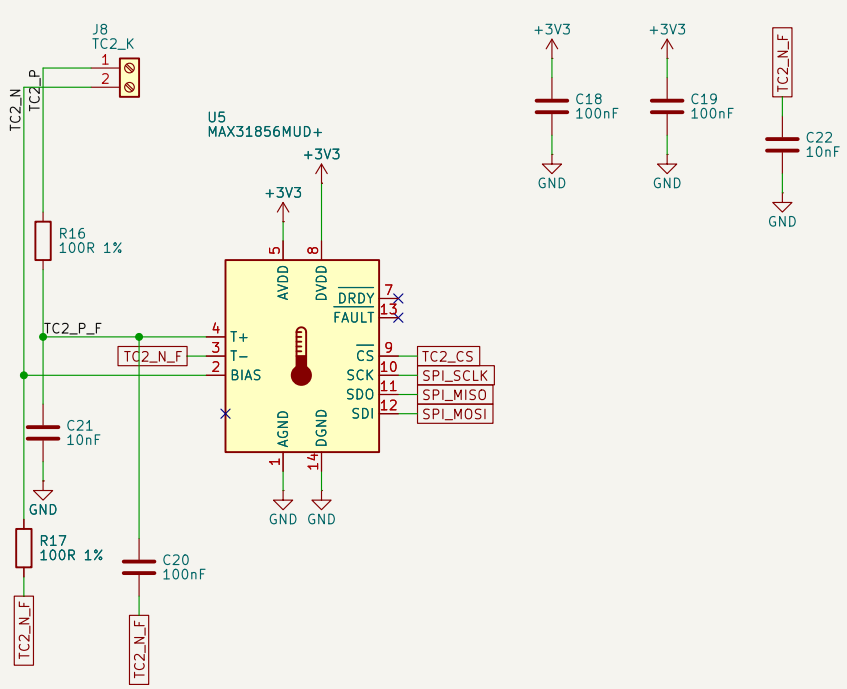
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



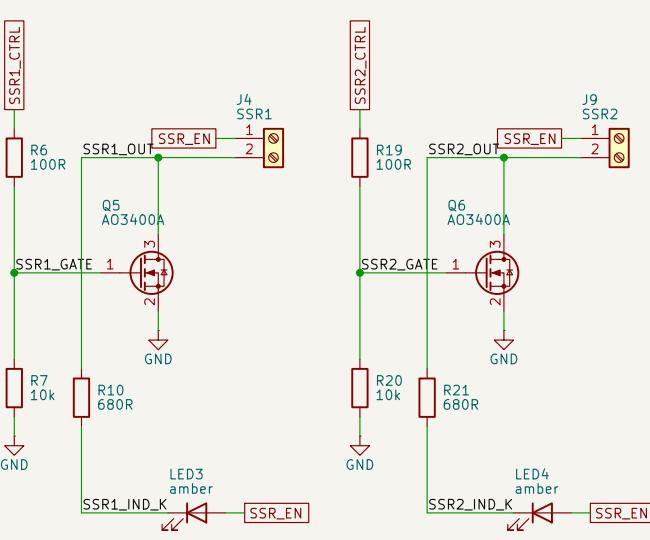
THERMOCOUPLE 1 (control) MAX31856
T- floats to J3, biased via BIAS



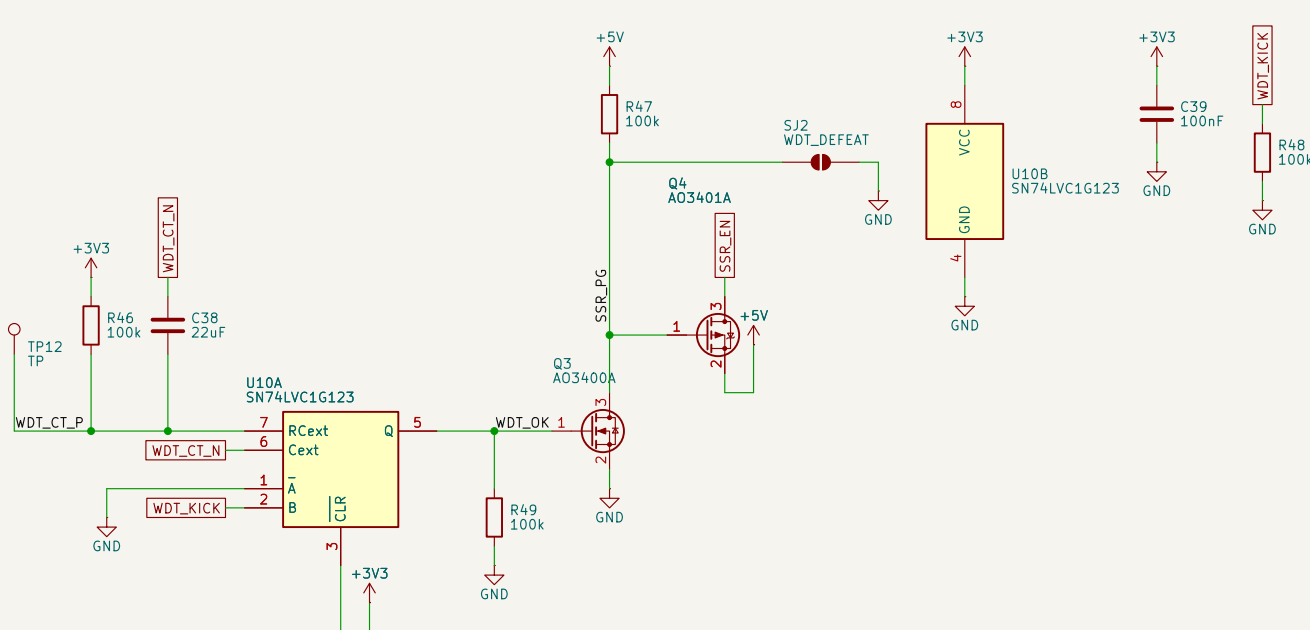
THERMOCOUPLE 2 (load) MAX31856
T- floats to J8, biased via BIAS



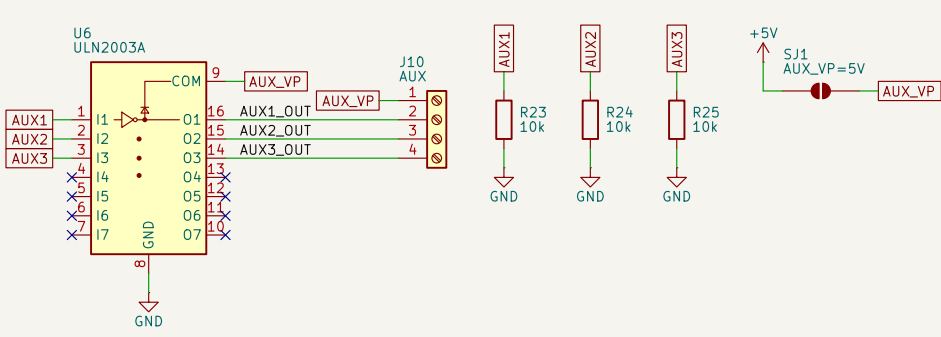
SSR DRIVE x2
low-side AO3400A; indicator LED across each terminal



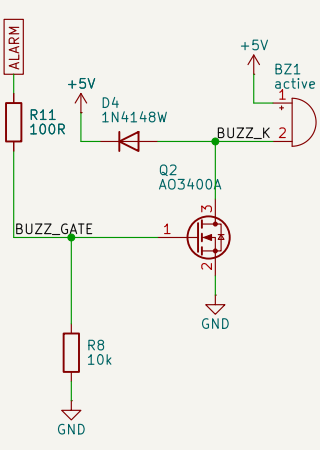
HARDWARE WATCHDOG
U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q3 stays high only while firmware keeps kicking. Q3 holds Q2 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN - the +5V rail feeding BOTH SSR terminals and both indicators.
R46/C38 set the window. R46 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.



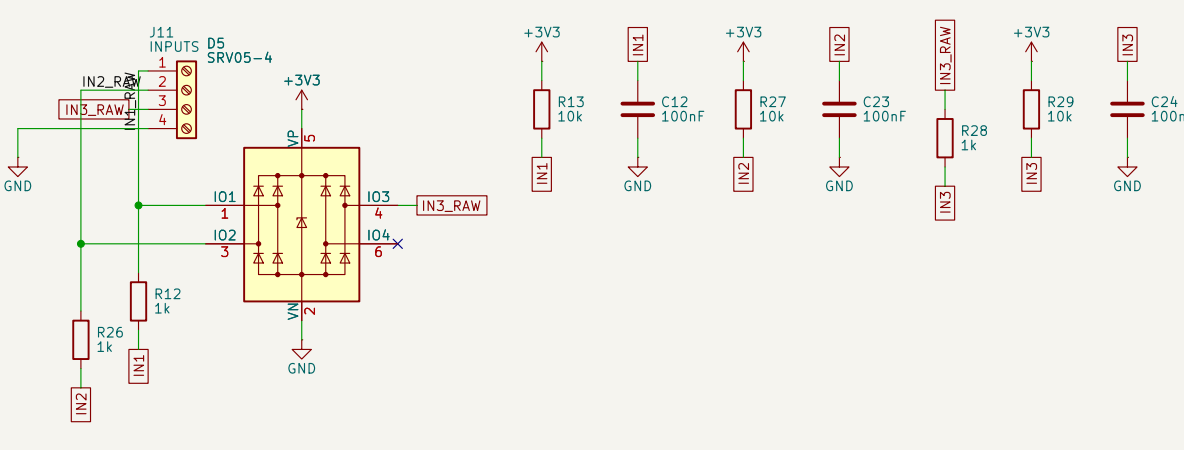
AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



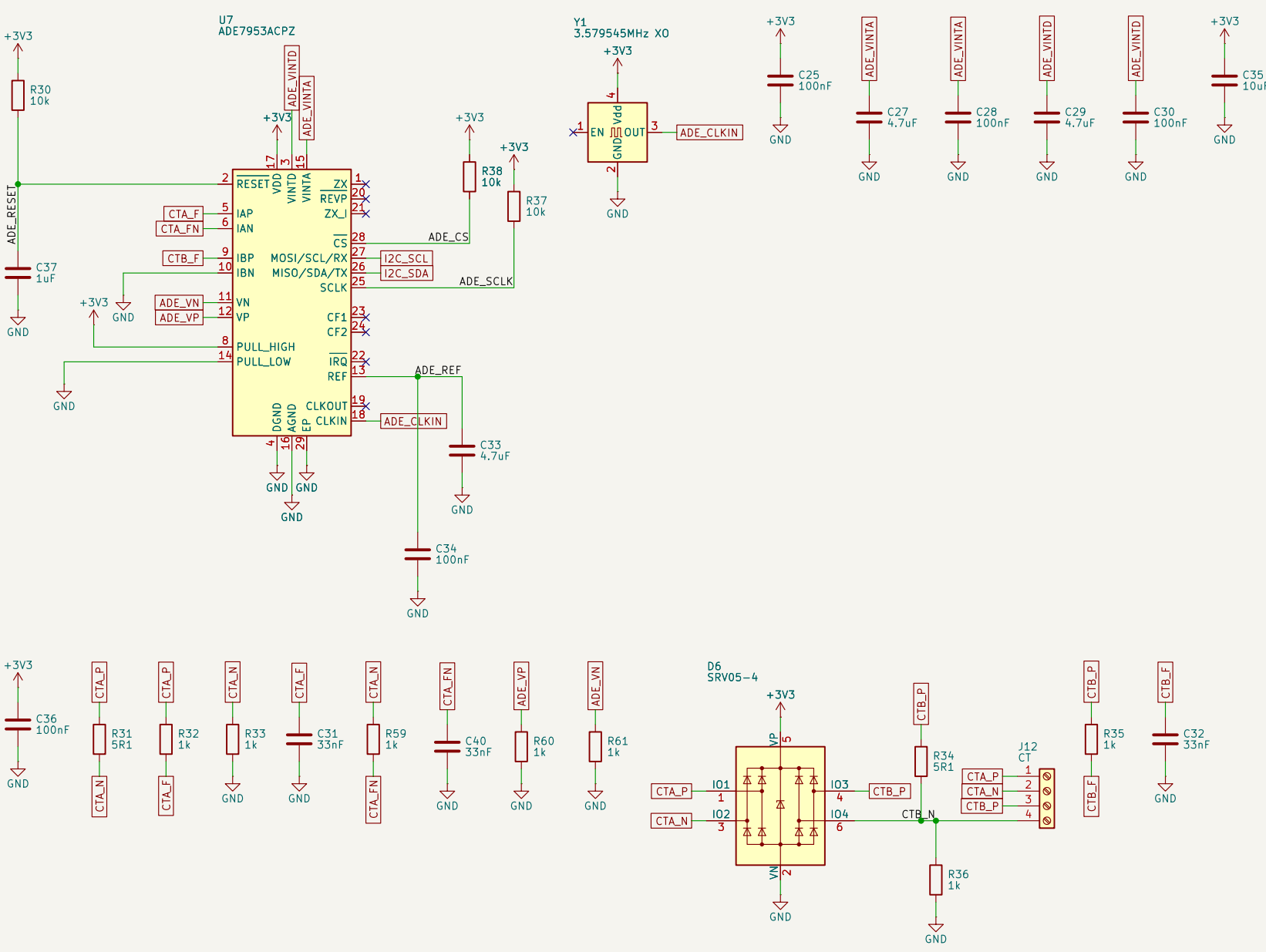
ALARM BUZZER



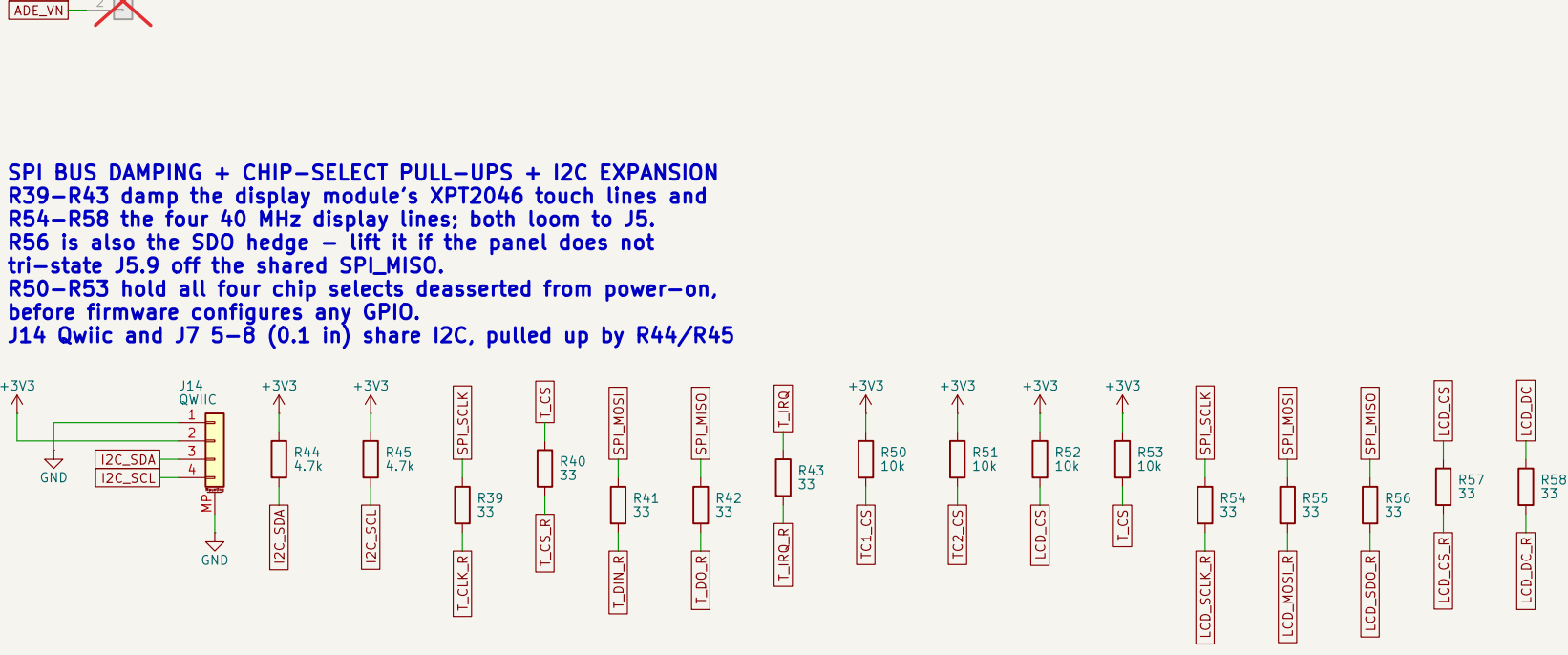
PROTECTED INPUTS x3: Iid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVs



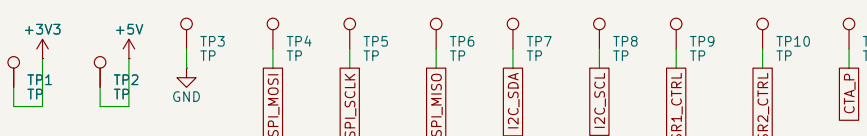
CT CURRENT SENSING: ADE7953, I2C, current-only
no mains - VP/VN to DNP J13



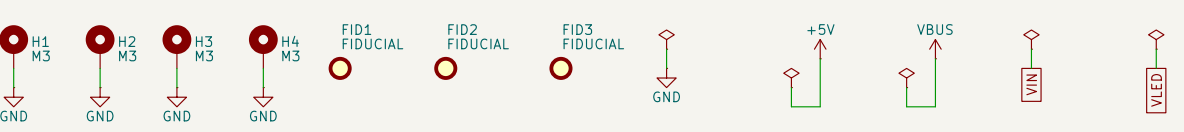
SPI BUS DAMPING + CHIP-SELECT PULL-UPS + I2C EXPANSION
R39-R43 damp the display module's XPT2046 touch lines and R54-R58 the four 40 MHz display lines; both loom to J5. R56 is also the SDO hedge - lift it if the panel does not tri-state J5.9 off the shared SPI_MISO. R50-R53 hold all four chip selects deasserted from power-on, before firmware configures any GPIO. J14 Qwic and J7 5-8 (0.1 in) share I2C, pulled up by R44/R45



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS
FID1-3 are bare-copper pick-and-place targets, on no net



NOTES - Bisque Kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated). pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated - rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.
TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal bias network. Ungrounded-junction probes required; two grounded-junction probes in one kiln would loop through it.
Nav switch J6 is panel-mounted: inputs use ESP32 pull-ups.
J11: IN1/IN2/IN3 (Iid/gas-flow/spare) + GND - dry contact each channel to GND.
Display J5 (14-pin, ST77965 + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SD0/MOSI 7=SSCK 8=BL 9=SD0/MISO 10=TDCLK 11=+V 12=+V 13=+V 14=+V 15=+V 16=+V 17=+V 18=+V 19=+V 20=+V 21=+V 22=+V 23=+V 24=+V 25=+V 26=+V 27=+V 28=+V 29=+V 30=+V 31=+V 32=+V 33=+V 34=+V 35=+V 36=+V 37=+V 38=+V 39=+V 40=+V 41=+V 42=+V 43=+V 44=+V 45=+V 46=+V 47=+V 48=+V 49=+V 50=+V 51=+V 52=+V 53=+V 54=+V 55=+V 56=+V 57=+V 58=+V 59=+V 60=+V 61=+V 62=+V 63=+V 64=+V 65=+V 66=+V 67=+V 68=+V 69=+V 70=+V 71=+V 72=+V 73=+V 74=+V 75=+V 76=+V 77=+V 78=+V 79=+V 80=+V 81=+V 82=+V 83=+V 84=+V 85=+V 86=+V 87=+V 88=+V 89=+V 90=+V 91=+V 92=+V 93=+V 94=+V 95=+V 96=+V 97=+V 98=+V 99=+V 100=+V 101=+V 102=+V 103=+V 104=+V 105=+V 106=+V 107=+V 108=+V 109=+V 110=+V 111=+V 112=+V 113=+V 114=+V 115=+V 116=+V 117=+V 118=+V 119=+V 120=+V 121=+V 122=+V 123=+V 124=+V 125=+V 126=+V 127=+V 128=+V 129=+V 130=+V 131=+V 132=+V 133=+V 134=+V 135=+V 136=+V 137=+V 138=+V 139=+V 140=+V 141=+V 142=+V 143=+V 144=+V 145=+V 146=+V 147=+V 148=+V 149=+V 150=+V 151=+V 152=+V 153=+V 154=+V 155=+V 156=+V 157=+V 158=+V 159=+V 160=+V 161=+V 162=+V 163=+V 164=+V 165=+V 166=+V 167=+V 168=+V 169=+V 170=+V 171=+V 172=+V 173=+V 174=+V 175=+V 176=+V 177=+V 178=+V 179=+V 180=+V 181=+V 182=+V 183=+V 184=+V 185=+V 186=+V 187=+V 188=+V 189=+V 190=+V 191=+V 192=+V 193=+V 194=+V 195=+V 196=+V 197=+V 198=+V 199=+V 200=+V 201=+V 202=+V 203=+V 204=+V 205=+V 206=+V 207=+V 208=+V 209=+V 210=+V 211=+V 212=+V 213=+V 214=+V 215=+V 216=+V 217=+V 218=+V 219=+V 220=+V 221=+V 222=+V 223=+V 224=+V 225=+V 226=+V 227=+V 228=+V 229=+V 230=+V 231=+V 232=+V 233=+V 234=+V 235=+V 236=+V 237=+V 238=+V 239=+V 240=+V 241=+V 242=+V 243=+V 244=+V 245=+V 246=+V 247=+V 248=+V 249=+V 250=+V 251=+V 252=+V 253=+V 254=+V 255=+V 256=+V 257=+V 258=+V 259=+V 260=+V 261=+V 262=+V 263=+V 264=+V 265=+V 266=+V 267=+V 268=+V 269=+V 270=+V 271=+V 272=+V 273=+V 274=+V 275=+V 276=+V 277=+V 278=+V 279=+V 280=+V 281=+V 282=+V 283=+V 284=+V 285=+V 286=+V 287=+V 288=+V 289=+V 290=+V 291=+V 292=+V 293=+V 294=+V 295=+V 296=+V 297=+V 298=+V 299=+V 300=+V 301=+V 302=+V 303=+V 304=+V 305=+V 306=+V 307=+V 308=+V 309=+V 310=+V 311=+V 312=+V 313=+V 314=+V 315=+V 316=+V 317=+V 318=+V 319=+V 320=+V 321=+V 322=+V 323=+V 324=+V 325=+V 326=+V 327=+V 328=+V 329=+V 330=+V 331=+V 332=+V 333=+V 334=+V 335=+V 336=+V 337=+V 338=+V 339=+V 340=+V 341=+V 342=+V 343=+V 344=+V 345=+V 346=+V 347=+V 348=+V 349=+V 350=+V 351=+V 352=+V 353=+V 354=+V 355=+V 356=+V 357=+V 358=+V 359=+V 360=+V 361=+V 362=+V 363=+V 364=+V 365=+V 366=+V 367=+V 368=+V 369=+V 370=+V 371=+V 372=+V 373=+V 374=+V 375=+V 376=+V 377=+V 378=+V 379=+V 380=+V 381=+V 382=+V 383=+V 384=+V 385=+V 386=+V 387=+V 388=+V 389=+V 390=+V 391=+V 392=+V 393=+V 394=+V 395=+V 396=+V 397=+V 398=+V 399=+V 400=+V 401=+V 402=+V 403=+V 404=+V 405=+V 406=+V 407=+V 408=+V 409=+V 410=+V 411=+V 412=+V 413=+V 414=+V 415=+V 416=+V 417=+V 418=+V 419=+V 420=+V 421=+V 422=+V 423=+V 424=+V 425=+V 426=+V 427=+V 428=+V 429=+V 430=+V 431=+V 432=+V 433=+V 434=+V 435=+V 436=+V 437=+V 438=+V 439=+V 440=+V 441=+V 442=+V 443=+V 444=+V 445=+V 446=+V 447=+V 448=+V 449=+V 450=+V 451=+V 452=+V 453=+V 454=+V 455=+V 456=+V 457=+V 458=+V 459=+V 460=+V 461=+V 462=+V 463=+V 464=+V 465=+V 466=+V 467=+V 468=+V 469=+V 470=+V 471=+V 472=+V 473=+V 474=+V 475=+V 476=+V 477=+V 478=+V 479=+V 480=+V 481=+V 482=+V 483=+V 484=+V 485=+V 486=+V 487=+V 488=+V 489=+V 490=+V 491=+V 492=+V 493=+V 494=+V 495=+V 496=+V 497=+V 498=+V 499=+V 500=+V 501=+V 502=+V 503=+V 504=+V 505=+V 506=+V 507=+V 508=+V 509=+V 510=+V 511=+V 512=+V 513=+V 514=+V 515=+V 516=+V 517=+V 518=+V 519=+V 520=+V 521=+V 522=+V 523=+V 524=+V 525=+V 526=+V 527=+V 528=+V 529=+V 530=+V 531=+V 532=+V 533=+V 534=+V 535=+V 536=+V 537=+V 538=+V 539=+V 540=+V 541=+V 542=+V 543=+V 544=+V 545=+V 546=+V 547=+V 548=+V 549=+V 550=+V 551=+V 552=+V 553=+V 554=+V 555=+V 556=+V 557=+V 558=+V 559=+V 560=+V 561=+V 562=+V 563=+V 564=+V 565=+V 566=+V 567=+V 568=+V 569=+V 570=+V 571=+V 572=+V 573=+V 574=+V 575=+V 576=+V 577=+V 578=+V 579=+V 580=+V 581=+V 582=+V 583=+V 584=+V 585=+V 586=+V 587=+V 588=+V 589=+V 590=+V 591=+V 592=+V 593=+V 594=+V 595=+V 596=+V 597=+V 598=+V 599=+V 600=+V 601=+V 602=+V 603=+V 604=+V 605=+V 606=+V 607=+V 608=+V 609=+V 610=+V 611=+V 612=+V 613=+V 614=+V 615=+V 616=+V 617=+V 618=+V 619=+V 620=+V 621=+V 622=+V 623=+V 624=+V 625=+V 626=+V 627=+V 628=+V 629=+V 630=+V 631=+V 632=+V 633=+V 634=+V 635=+V 636=+V 637=+V 638=+V 639=+V 640=+V 641=+V 642=+V 643=+V 644=+V 645=+V 646=+V 647=+V 648=+V 649=+V 650=+V 651=+V 652=+V 653=+V 654=+V 655=+V 656=+V 657=+V 658=+V 659=+V 660=+V 661=+V 662=+V 663=+V 664=+V 665=+V 666=+V 667=+V 668=+V 669=+V 670=+V 671=+V 672=+V 673=+V 674=+V 675=+V 676=+V 677=+V 678=+V 679=+V 680=+V 681=+V 682=+V 683=+V 684=+V 685=+V 686=+V 687=+V 688=+V 689=+V 690=+V 691=+V 692=+V 693=+V 694=+V 695=+V 696=+V 697=+V 698=+V 699=+V 700=+V 701=+V 702=+V 703=+V 704=+V 705=+V 706=+V 707=+V 708=+V 709=+V 710=+V 711=+V 712=+V 713=+V 714=+V 715=+V 716=+V 717=+V 718=+V 719=+V 720=+V 721=+V 722=+V 723=+V 724=+V 725=+V 726=+V 727=+V 728=+V 729=+V 730=+V 731=+V 732=+V 733=+V 734=+V 735=+V 736=+V 737=+V 738=+V 739=+V 740=+V 741=+V 742=+V 743=+V 744=+V 745=+V 746=+V 747=+V 748=+V 749=+V 750=+V 751=+V 752=+V 753=+V 754=+V 755=+V 756=+V 757=+V 758=+V 759=+V 760=+V 761=+V 762=+V 763=+V 764=+V 765=+V 766=+V 767=+V 768=+V 769=+V 770=+V 771=+V 772=+V 773=+V 774=+V 775=+V 776=+V 777=+V 778=+V 779=+V 780=+V 781=+V 782=+V 783=+V 784=+V 785=+V 786=+V 787=+V 788=+V 789=+V 790=+V 791=+V 792=+V 793=+V 794=+V 795=+V 796=+V 797=+V 798=+V 799=+V 800=+V 801=+V 802=+V 803=+V 804=+V 805=+V 806=+V 807=+V 808=+V 809=+V 810=+V 811=+V 812=+V 813=+V 814=+V 815=+V 816=+V 817=+V 818=+V 819=+V 820=+V 821=+V 822=+V 823=+V 824=+V 825=+V 826=+V 827=+V 828=+V 829=+V 830=+V 831=+V 832=+V 833=+V 834=+V 835=+V 836=+V 837=+V 838=+V 839=+V 840=+V 841=+V 842=+V 843=+V 844=+V 845=+V 846=+V 847=+V 848=+V 849=+V 850=+V 851=+V 852=+V 853=+V 854=+V 855=+V 856=+V 857=+V 858=+V 859=+V 860=+V 861=+V 862=+V 863=+V 864=+V 865=+V 866=+V 867=+V 868=+V 869=+V 870=+V 871=+V 872=+V 873=+V 874=+V 875=+V 876=+V 877=+V 878=+V 879=+V 880=+V 881=+V 882=+V 883=+V 884=+V 885=+V 886=+V 887=+V 888=+V 889=+V 890=+V 891=+V 892=+V 893=+V 894=+V 895=+V 896=+V 897=+V 898=+V 899=+V 900=+V 901=+V 902=+V 903=+V 904=+V 905=+V 906=+V 907=+V 908=+V 909=+V 910=+V 911=+V 912=+V 913=+V 914=+V 915=+V 916=+V 917=+V 918=+V 919=+V 920=+V 921=+V 922=+V 923=+V 924=+V 925=+V 926=+V 927=+V 928=+V 929=+V 930=+V 931=+V 932=+V 933=+V 934=+V 935=+V 936=+V 937=+V 938=+V 939=+V 940=+V 941=+V 942=+V 943=+V 944=+V 945=+V 946=+V 947=+V 948=+V 949=+V 950=+V 951=+V 952=+V 953=+V 954=+V 955=+V 956=+V 957=+V 958=+V 959=+V 960=+V 961=+V 962=+V 963=+V 964=+V 965=+V 966=+V 967=+V 968=+V 969=+V 970=+V 971=+V 972=+V 973=+V 974=+V 975=+V 976=+V 977=+V 978=+V 979=+V 980=+V 981=+V 982=+V 983=+V 984=+V 985=+V 986=+V 987=+V 988=+V 989=+V 990=+V 991=+V 992=+V 993=+V 994=+V 995=+V 996=+V 997=+V 998=+V 999=+V 1000=+V 1001=+V 1002=+V 1003=+V 1004=+V 1005=+V 1006=+V 1007=+V 1008=+V 1009=+V 1010=+V 1011=+V 1012=+V 1013=+V 1014=+V 1015=+V 1016=+V 1017=+V 1018=+V 1019=+V 1020=+V 1021=+V 1022=+V 1023=+V 1024=+V 1025=+V 1026=+V 1027=+V 1028=+V 1029=+V 1030=+V 1031=+V 1032=+V 1033=+V 1034=+V 1035=+V 1036=+V 1037=+V 1038=+V 1039=+V 1040=+V 1041=+V 1042=+V 1043=+V 1044=+V 1045=+V 1046=+V 1047=+V 1048=+V 1049=+V 1050=+V 1051=+V 1052=+V 1053=+V 1054=+V 1055=+V 1056=+V 1057=+V 1058=+V 1059=+V 1060=+V 1061=+V 1062=+V 1063=+V 1064=+V 1065=+V 1066=+V 1067=+V 1068=+V 1069=+V 1070=+V 1071=+V 1072=+V 1073=+V 1074=+V 1075=+V 1076=+V 1077=+V 1078=+V 1079=+V 1080=+V 1081=+V 1082=+V 1083=+V 1084=+V 1085=+V 1086=+V 1087=+V 1088=+V 1089=+V 1090=+V 1091=+V 1092=+V 1093=+V 1094=+V 1095=+V 1096=+V 1097=+V 1098=+V 1099=+V 1100=+V 1101=+V 1102=+V 1103=+V 1104=+V 1105=+V 1106=+V 1107=+V 1108=+V 1109=+V 1110=+V 1111=+V 1112=+V 1113=+V 1114=+V 1115=+V 1116=+V 1117=+V 1118=+V 1119=+V 1120=+V 1121=+V 1122=+V 1123=+V 1124=+V 1125=+V 1126=+V 1127=+V 1128=+V 1129=+V 1130=+V 1131=+V 1132=+V 1133=+V 1134=+V 1135=+V 1136=+V 1137=+V 1138=+V 1139=+V 1140=+V 1141=+V 1142=+V 1143=+V 1144=+V 1145=+V 1146=+V 1147=+V 1148=+V 1149=+V 1150=+V 1151=+V 1152=+V 1153=+V 1154=+V 1155=+V 1156=+V 1157=+V 1158=+V 1159=+V 1160=+V 1161=+V 1162=+V 1163=+V 1164=+V 1165=+V 1